



A processing-based explanation for left-right agreement asymmetries

Omri Amiraz

The Hebrew University of Jerusalem

Overview

- This paper investigates cases where the presence of agreement depends on the relative linear order of the agreement controller and target.
- The best-studied type of left-right agreement asymmetry is in bound verbal subject marking.
- However, such asymmetries are not restricted to argument indexing.
- I propose a processing-based explanation in the spirit of Dryer (1992) and Hawkins (1994):
 - Agreement incurs a processing cost in constructions where the target precedes the controller because it forces the speaker to plan further ahead than would be otherwise required.
 - Consequently, there is a functional pressure to neutralize agreement in such constructions.

Asymmetries in verbal subject marking

Romance languages:

In Fiorentino and Trentino, the verb shows default agreement in VS clauses (Brandi and Cordin, 1989):

- (1) È vegnú qualche putela.
be.3SG come.PTCP some girls
‘Some girls have come.’ (Trentino)

In certain varieties of Brazilian Portuguese, number is only optionally neutralized in VS clauses (Guasti and Rizzi, 2002):

- (2) {Chegaram / chegou} dois meninos.
came.PL came.SG two boys
‘Two boys came.’

Bantu languages:

Many Bantu languages have default agreement in subject inversion constructions (Marten and Van der Wal, 2014):

- (3) P-è-yá òvá-éndà.
SM16-PST-come 2CC-visitor
‘Visitors came.’ (Herero)

Semitic languages:

In Standard Arabic, the verb shows default singular agreement in VS clauses (Bolotin, 1995):

- (4) {*nimna / naamat} al-banaat-u.
slept.3PL.F slept.3SG.F the-girls-NOM
‘The girls slept.’

Empirical generalization

I propose the following cross-linguistic generalization:

- (5) Left-right agreement asymmetry generalization:
Agreement is more likely to be partially/fully neutralized when the target precedes the controller than when the controller precedes the target.

This generalization is somewhat similar to the Direction of Agreement Parameter proposed by Baker (2008):

- (6) The Direction of Agreement Parameter:
F agrees with DP/NP only if DP/NP asymmetrically c-commands F (Baker, 2008, 155).

However, there are two major differences:

- The generalization in (5) is statistical rather than absolute, i.e., it allows for cases where agreement neutralization is optional.
- The generalization in (5) allows for cases where agreement neutralization is restricted to particular constructions rather than happening across the board.

Other left-right agreement asymmetries

Left-right agreement asymmetries are not restricted to argument indexing.

In Hebrew, agreement asymmetries are found in a variety of constructions:

- Bound verbal subject marking
- The prenominal indefinite determiner optionally agrees with the head noun in number and gender (7a). In contrast, agreement is obligatory in the case of postnominal indefinite determiners (7b).

- (7) a. hu gar {be-ezo / be-eze} ir
he live in-some.SG.F in-some.SG.M city(F)
be-italia.
in-Italy
‘He lives in some city in Italy.’
b. hu gar be-ir {kolšehi / *kolšehu}
he live in-city(F) some.SG.F some.SG.M
be-italia.
in-Italy
‘He lives in some city in Italy.’

- Numerals generally precede the noun. Some speakers use the feminine numeral form, which is shorter than the masculine form, in some or all cases (8a). However, this is not possible with the numeral *axat* ‘one’, which follows the noun, unlike the other numerals (8b). Yet, in a partitive construction, *axat* ‘one’ precedes the noun, and agreement is again optional (8c).

- (8) a. {šloša / šaloš} maki-ot
three.M three.F stick(M)-PL
‘Three sticks’
b. makel {exad / *axat}
stick(M) one.M one.F
‘One stick’
c. {axat / exad} ha-sib-ot
one.of.F one.of.M DEF-reason(F)-PL
‘One of the reasons’

- The intensifier *kaze* ‘such’ allows default agreement when preceding the noun.

- (9) a. ze {kazot / kaze} beaya
it such.F such.M problem(F)
meachen-et.
annoying-SG.F
‘It’s such an annoying problem.’
b. ze beaya {kazot / ?kaze}
it problem(F) such.F such.M
meachen-et.
annoying-SG.F
‘It’s such an annoying problem.’

A processing-based explanation

The generalization in (5) cannot be subsumed under differential argument indexing and requires an explanation general enough to apply to non-argument agreement.

It is a case of what Haspelmath (2019) calls multi-convergence: a situation where a similar result comes about through diverse diachronic pathways.

I propose a processing-based explanation in the spirit of Dryer (1992) and Hawkins (1994):

- Agreement incurs a processing cost in constructions where the target precedes the controller because it forces the speaker to plan further ahead than would be otherwise required.
- Consequently, there is a functional pressure to neutralize agreement in such constructions.

Speaker vs. listener-oriented efficiency

A language is efficient if the speaker and listener can successfully communicate with minimal cognitive, articulatory, and perceptual effort on average (Gibson et al., 2019).

From this perspective, a fundamental question arises about the evolution of grammar: do languages tend to favor efficiency for the speaker or the listener?

Syntactic agreement is an intriguing case study because these two criteria of efficiency appear to be in conflict.

Agreement comprehension (the listener)

Some scholars have argued that agreement increases efficiency in that it facilitates comprehension by providing a cue about the identity of the controller (Gibson et al., 2019).

Agreement thus facilitates comprehension in those constructions where the target precedes the controller.

Agreement production (the speaker)

Previous studies show that when the subject (controller) precedes the verb (target), speakers can begin to speak without planning the verb in advance, which reduces memory demands (Momma et al., 2016).

But what happens when it is the target that precedes the controller?

I argue that agreement forces the speaker to plan the controller beforehand in order to retrieve its features from memory and choose the correct inflectional form of the target.

This likely incurs a processing cost by increasing the memory load.

Hence, when the target precedes the controller, agreement provides a cue for the listener but requires extra effort on the part of the speaker.

How agreement asymmetries emerge

The generalization in (5) supports the hypothesis that agreement systems evolve over time to increase efficiency for the speaker at the expense of the listener.

I further hypothesize that agreement asymmetries arise from two diachronic tendencies:

- Languages are more likely to lose their agreement morphology in constructions where there is greater reduction of processing difficulty for the speaker, i.e., when the target precedes the controller.
- Constructions that neutralize agreement are more likely to spread to novel contexts if they facilitate production.

References

- Baker, M. C. (2008). The syntax of agreement and concord. Cambridge: Cambridge University Press.
- Bolotin, N. (1995). Arabic and parametric VSO agreement. In M. Eid (Ed.), Perspectives on Arabic linguistics VII: Papers from the seventh Annual Symposium on Arabic Linguistics, pp. 7–27. John Benjamins.
- Brandi, L. and P. Cordin (1989). Two Italian dialects and the null subject parameter. In O. Jaeggli and K. J. Safir (Eds.), The null subject parameter, pp. 111–142. Kluwer Academic Publishers.
- Dryer, M. S. (1992). The Greenbergian word order correlations. Language 68(1), 81–138.
- Gibson, E., R. Futrell, S. P. Piantadosi, I. Dautriche, K. Mahowald, L. Bergen, and R. Levy (2019). How efficiency shapes human language. Trends in Cognitive Sciences 23(5), 389–407.
- Guasti, M. T. and L. Rizzi (2002). Agreement and tense as distinct syntactic positions: Evidence from acquisition. In G. Cinque (Ed.), Functional structure in DP and IP, pp. 167–194. Oxford University Press New York.
- Haspelmath, M. (2019). Can cross-linguistic regularities be explained by constraints on change? In K. Schmidtke-Bode, N. Levshina, S. M. Michaelis, and I. Seržant (Eds.), Explanation in typology: Diachronic sources, functional motivations and the nature of the evidence, pp. 1–23. Berlin: Language Science Press.
- Hawkins, J. A. (1994). A performance theory of order and constituency. Cambridge: Cambridge University Press.
- Marten, L. and J. Van der Wal (2014). A typology of Bantu subject inversion. Linguistic Variation 14(2), 318–368.
- Momma, S., L. R. Slevc, and C. Phillips (2016). The timing of verb selection in Japanese sentence production. Journal of Experimental Psychology: Learning, Memory, and Cognition 42(5), 813–824.